EF101-P04: Computer Hardware Stress Testing and Resource Analysis

Submission Date: 9 November 2026
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1. Mandatory Experimental Setup:-

Test-Machine Specifications:-

- Laptop/Desktop Form Factor: [Laptop]
- Operating System: [Windows 11 Home]
- CPU Model: [Intel Core i7-1165G7]
- Core/Thread Count: [4 Cores / 8 Threads]
- Installed RAM: [12 GB DDR4]
- RAM Capacity used by OS: [11.8 GB usable physical RAM]
- GPU Model: Intel Iris Xe Graphics (Integrated)
- Storage Type: NVMe KINGSTON OM8PCP3512F-AA

Tools Used:-

- Monitoring Tools: HWiNFO64 (used to automatically log CPU/RAM/GPU utilization, temperatures, and disk throughput to a CSV file every [approximately 2] seconds) and PresentMon (for framerate).
- Stress-Testing Tools: HeavyLoad was used for stress-testing of CPU, GPU, RAM, and SSD.

2. Required Analysis

A. Idle vs. Stress Comparisons:-

(See the images/ folder for all generated graphs)

As shown in the graph above, we established an idle baseline for each component before applying the stress test. A brief overview of each component is given below as well.

CPU: The total CPU usage in idle state was nearly 21%, during the stress test it reached the maximum point of 100% with core temperature reaching 88°C.
GPU: During idle conditions, GPU usage was minimal (around 2-5%) with low temperatures around 40°C. Under stress testing, GPU utilization spiked to 100%, causing temperatures to rise rapidly into the low 90s (°C) and triggering severe clock speed reductions down to ~600 MHz to manage the heat.
Storage: In the idle state, disk activity was effectively 0%. During testing, storage active time reached 100% continuous utilization with data transfer speeds reaching up to 692 MB/s, while temperatures remained stable.
RAM: Baseline idle RAM utilization sat around 3.5 GB (roughly 30%). When subjected to stress testing and heavy multitasking workloads, memory consumption increased

significantly to ~9.5 GB (80% capacity), forcing the operating system to increase virtual memory page file usage on the SSD to handle the extra load.

B. Hardware Bottlenecks:-

Based on our collected data, the limiting resource (bottleneck) for each workload was:

CPU Test: The CPU (100% usage).
Storage Test: The SSD (100% active time).
GPU Test: The GPU (100% usage).
Mixed Workload (Tabs + Video + Zipping): The SSD was the primary bottleneck at 86% average active time (frequently spiking to 100%), followed by RAM at 77% and CPU at 72%.

C. Utilization vs. Performance:-

Our data highlights a critical difference between utilization and performance. During the CPU test, CPU usage was 100% meaning that it was being used all of the time. However, the actual utility (work done relative to base speed) fluctuated around 49%. This shows that a component can be 100% utilized, but if it is running hot and at a slowed clock speed, it is not performing at 100% of its maximum capacity.

D. RAM Pressure and Virtual Memory:-

During our mixed workload, physical RAM reached around 80% capacity (9.5 GB used out of 11.8 GB). As RAM pressure increased, the operating system utilized virtual memory to prevent crashes, evidenced by the page file on the SSD growing from 1.77 GB to 2.02 GB. The OS moves lower-priority background data from physical RAM to this slower SSD page file to free up space for active tasks.

E. Clock Changes, Thermal Throttling, and Power Limits:-

CPU: During the CPU stress test, the package temperature sat nearly around 90°C. Upon reaching 100% load, thermal throttling engaged immediately, causing the clock speed to drop from ~3000 MHz to between 1000-2000 MHz (well below the 2800 MHz, which is the base speed).
GPU: During the GPU stress test, temperatures reached the low 90s (°C). This sustained load caused the clock speed to throttle heavily, mostly hovering around 600 MHz and causing the desktop frame rate to plummet to nearly 2 FPS.

F. Comparing Two Different Stress Tasks:-

When comparing the CPU stress test and the Storage stress test, the system behaves very differently. The CPU test relies entirely on processing power, generating massive heat and triggering thermal throttling, but requires very little data transfer from the drive. Conversely, the Storage test requires moving data at up to 692 MB/s, pinning the SSD's active time at

100%, but only required 65% RAM usage and did not heavily tax the CPU's compute capabilities.

G. Why a System Feels Slow Below 100% CPU:-

A system can feel incredibly sluggish even if the CPU is not at 100%. This is usually due to a storage bottleneck. As seen in our storage test, the SSD active time was at 100% even when moving only 1-20 MB/s of small files. When the drive is 100% busy fielding a queue of requests, the CPU has to wait for data to arrive, making the entire user interface feel unresponsive despite the CPU having plenty of available processing power.

3. Submission Evidence & Limitations:-

Test Durations: Isolated tests were run for roughly [For CPU, storage and GPU Tests were approximately 65-110] seconds each, and the mixed workload was logged for over 150 seconds.
Raw Data: All raw logs and cleaned CSV data can be found in the /data/ directory of this repository.
Limitations: All tests were conducted while the laptop was running on battery power and the system was already operating at high temperatures (~90°C). This elevated starting heat caused thermal throttling right at the start of testing, slowing down processor performance and preventing the capture of peak non-throttled performance baseline data.